

LocalLife Waste Monitoring System

Technical Architecture & Process Pipeline Overview

System Overview

The **LocalLife Waste Monitoring System** automatically measures garbage bin fill levels and identifies waste categories in real time. Positioned directly above a fixed bin, the setup utilizes a dual-camera configuration connected to a local Raspberry Pi and processing host. The entire system operates **100% offline** without requiring cloud connectivity.

- **Hardware:** Intel RealSense D435 (Depth + RGB), Logitech C920 (RGB), Raspberry Pi, Windows host processing unit.
- **Primary Output:** Real-time volume estimation (Liters), material classification, object count, and color categorization.

System Architecture & Pipeline Phases

Step 1: Multi-Modal Data Capture

The system ingests video streams simultaneously from both sensors at approximately 30 frames per second (fps):

- **Intel RealSense D435:** Synchronized capture of aligned RGB photos and 16-bit depth maps (per-pixel physical distance metrics).
- **Logitech C920:** High-resolution optical RGB stream for detailed surface visual inspection.

Step 2: Object Detection (YOLOv8)

Frame-by-frame detection is executed via **Ultralytics YOLOv8** to identify distinct items inside the bin perimeter:

- **Bounding Box Coordinates:** Spatial location of items (garbage bags, cardboard boxes, loose plastic).
- **Classification Confidence:** Quantitative certainty score (0–100%) per item instance.

Step 3: Material & Color Classification (OpenAI CLIP)

Image crops of detected items are passed to an **OpenAI CLIP** vision-language model (HuggingFace Transformers library) to evaluate semantic attributes:

- **Material Types:** Distinguishes plastic, fabric, cardboard, organic, and composite materials.
- **Color Profiling:** Predicts primary colors (e.g., black garbage bag vs. brown cardboard box).

Step 4: Depth Processing & 3D Volume Calculation

Raw depth maps are transformed into volumetric metrics through spatial geometry algorithms:

- **Background Subtraction:** Filters out coordinates residing outside the internal bin boundaries.
- **Intelligent Hole Filling:** Neighbor-based iterative reconstruction for depth shadows or non-reflective dark surfaces.
- **Volumetric Integration:** Translates depth pixel offsets into 3D volume (Liters) using camera intrinsic matrix parameters and frustum geometry.

Step 5: Multi-Frame Object Tracking & Phantom Rejection

A custom tracking algorithm ensures temporal consistency across sequential frames:

- **Entity Persistence:** Assigns unique IDs to detected objects to prevent duplicate counting as items persist in the bin (~24 frame lifespan validation).
- **Phantom Removal:** Filters out transient depth artifacts or shadow noise (which typically dissolve within ~2 frames) without neural model confirmation.

Step 6: Dual-Camera Data Fusion

To eliminate conflicting sensor outputs, a sensor fusion layer merges inputs based on sensor strengths:

- **Depth Priority:** RealSense provides the primary volumetric payload.
- **Visual Priority:** Logitech provides higher-fidelity color profiles, paired with CLIP material inferences.
- **Output:** Unifies separate data streams into a single consolidated reading on the local Flask web dashboard.

Step 7: Reference Calibration & Error Correction

In situ spatial variances are adjusted using a baseline calibration routine:

- A known physical volume (e.g., a 10-Liter reference object) is measured by the system.
- A scaling calibration offset ratio is calculated and stored.
- Subsequent volume calculations dynamically apply this correction factor.

Core Technologies & Dependency Matrix

Component / Subsystem	Library / Technology	Functional Role
Object Detection	Ultralytics (YOLOv8)	Real-time bounding box detection and preliminary item labeling.
Semantic Classification	OpenAI CLIP / Transformers	Zero-shot material composition and color identification.
Depth & Spatial Math	NumPy, OpenCV	Array manipulation, background clipping, 3D point cloud volume calculation.
Concurrency Control	Python threading / asyncio	Parallel asynchronous ingestion of dual high-fps camera feeds.
Hardware Interface	Intel RealSense SDK (pyrealsense2)	Hardware communication, depth alignment, and IR stream access.
Dashboard & UI	Flask, HTML5, JavaScript	Local web console presenting live metrics, system state, and fill levels.

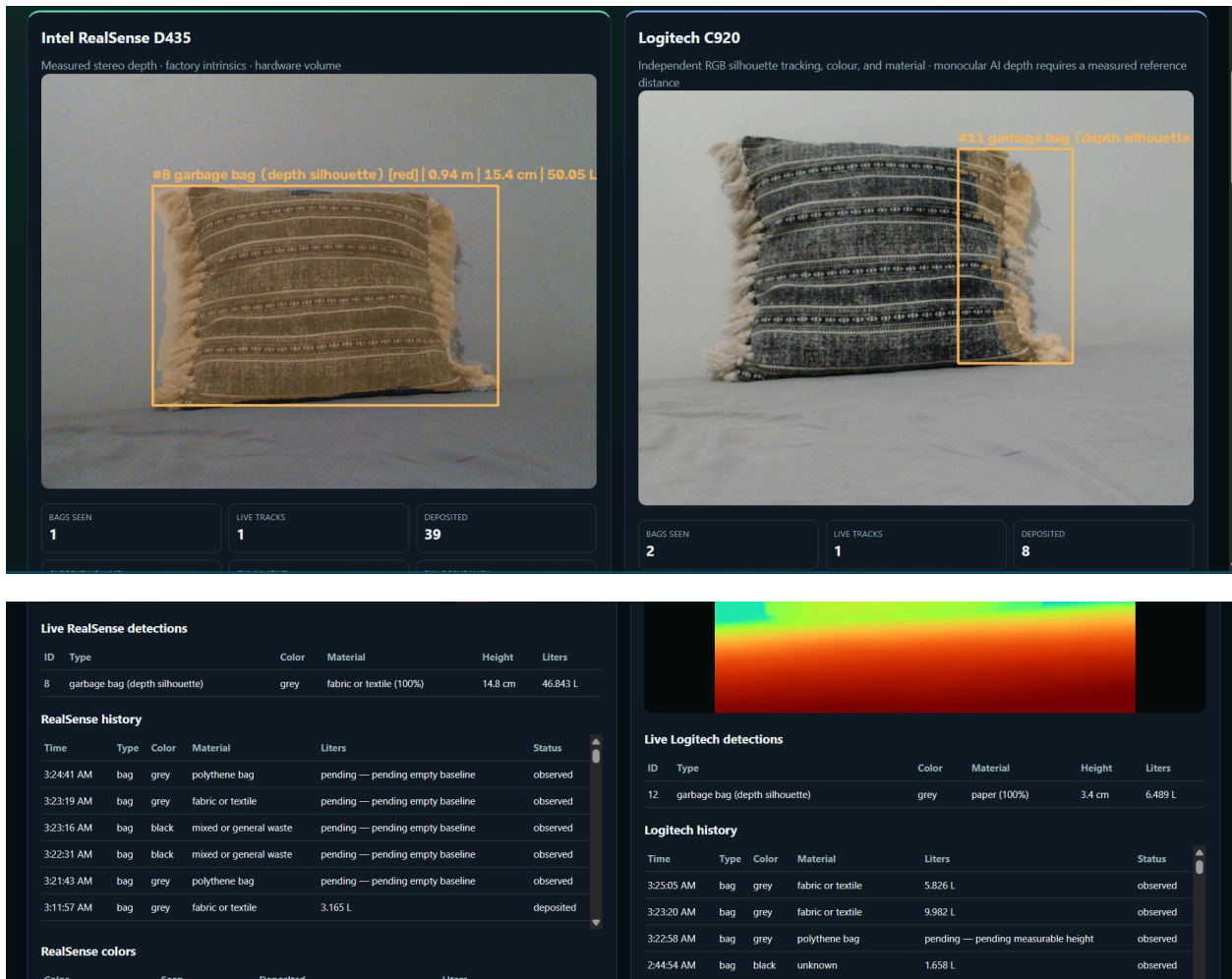
System Execution Flow

1. System Initialization & Hardware Initialization.
2. Asynchronous frame ingestion from RealSense D435 and Logitech C920.
3. Pass RGB frame through YOLOv8 for item detection.
4. Crop detected regions and run OpenAI CLIP for material/color tags.
5. Fetch aligned depth matrix from RealSense.
6. Execute spatial filtering: crop background & execute neighbor-fill on invalid depth pixels.
7. Compute 3D volume using calibrated camera intrinsic parameters.
8. Evaluate object tracker: update frame persistence counter & filter transient phantom artifacts.

- 9. Perform sensor fusion (RealSense depth + Logitech color + CLIP tags).
- 10. Update Flask live dashboard stream.
- 11. Repeat loop for next operational frame.

Key Architectural Advantages

- **Sensor Synergy:** Fuses structural depth sensing with high-definition optical clarity, mitigating standalone sensor weaknesses.
- **Robust Object Intelligence:** Leverages deep neural networks (YOLO + CLIP) rather than naive pixel-matching heuristics.
- **Temporal Tracking:** Prevents overcounting static items sitting in view over continuous frame sequences.
- **Offline Privacy & Reliability:** Eliminates cloud dependency, ensuring zero latency, zero subscription costs, and total data privacy.



Only logitech currently has a confirmed object showing its measurement alone

Intel RealSense D435

Measured stereo depth - factory intrinsics - hardware volume



BAGS SEEN
2

LIVE TRACKS
1

DEPOSITED
39

Logitech C920

Independent RGB silhouette tracking, colour, and material - monocular AI depth requires a measured reference distance



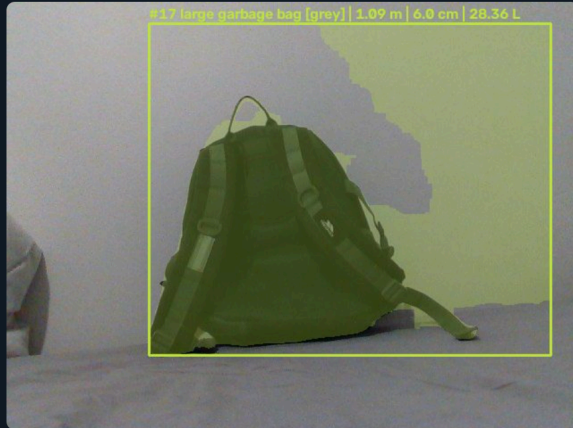
BAGS SEEN
1

LIVE TRACKS
1

DEPOSITED
10

Intel RealSense D435

Measured stereo depth - factory intrinsics - hardware volume



BAGS SEEN
8

LIVE TRACKS
1

DEPOSITED
42

CURRENT VOLUME
38.357 L

CUMULATIVE
563.077 L

BIN OCCUPANCY
38.358 L

Logitech C920

Independent RGB silhouette tracking, colour, and material - monocular AI depth requires a measured reference distance



BAGS SEEN
25

LIVE TRACKS
1

DEPOSITED
10

CURRENT VOLUME
38.357 L

CUMULATIVE
563.077 L

BIN OCCUPANCY
38.358 L

#20 garbage bag (depth silhouette)



BAGS SEEN
1

LIVE TRACKS
1

DEPOSITED
42

CURRENT VOLUME
12.812 L

CUMULATIVE
562.877 L

BIN OCCUPANCY
12.833 L

Capture baseline nowReset baselineDownload RealSense CSV

distance



BAGS SEEN
1

LIVE TRACKS
0

DEPOSITED
10

CURRENT VOLUME
—

CUMULATIVE
583.020 L

BIN OCCUPANCY
—

Intel RealSense D435

Measured stereo depth · factory intrinsics · hardware volume

#20 garbage bag (depth silhouette) [grey] | 1.00 m | 11.7 cm | 3



BAGS SEEN
1

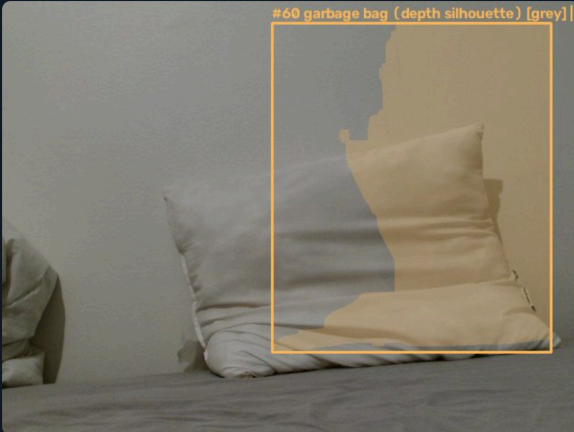
LIVE TRACKS
1

DEPOSITED
42

Logitech C920

Independent RGB silhouette tracking, colour, and material · monocular AI depth requires a measured reference distance

#60 garbage bag (depth silhouette) [grey] |



BAGS SEEN
4

LIVE TRACKS
1

DEPOSITED
10

Live RealSense detections

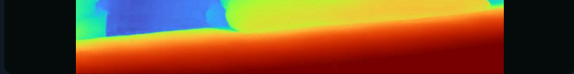
ID	Type	Color	Material	Height	Liters
20	garbage bag (depth silhouette)	grey	polythene bag (75%)	11.6 cm	34.349 L

RealSense history

Time	Type	Color	Material	Liters	Status
3:34:59 AM	bag	grey	paper	3.493 L	deposited
3:34:23 AM	bag	black	cardboard	27.564 L	deposited
3:33:51 AM	bag	grey	paper	3.964 L	observed
3:31:34 AM	bag	brown	fabric or textile	85.330 L	deposited
3:24:41 AM	bag	grey	polythene bag	pending — pending empty baseline	observed
3:23:19 AM	bag	grey	fabric or textile	pending — pending empty baseline	observed

RealSense colors

Color	Seen	Deposited	Liters
grey	1	26	338.507 L
black	0	9	46.164 L
brown	0	5	175.096 L



Live Logitech detections

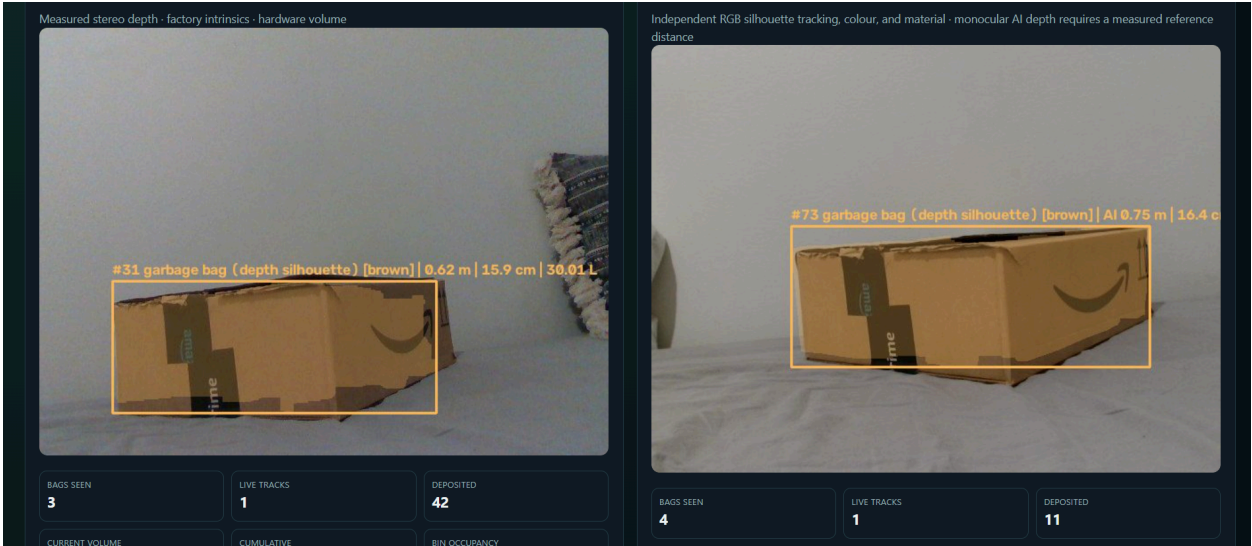
ID	Type	Color	Material	Height	Liters
60	garbage bag (depth silhouette)	grey	fabric or textile (100%)	12.8 cm	39.078 L

Logitech history

Time	Type	Color	Material	Liters	Status
3:37:19 AM	bag	grey	polythene bag	43.172 L	observed
3:37:01 AM	bag	grey	polythene bag	10.914 L	observed
3:35:00 AM	bag	grey	paper	8.547 L	deposited
3:34:49 AM	bag	grey	polythene bag	6.500 L	observed
3:33:22 AM	bag	grey	fabric or textile	50.569 L	observed
3:32:17 AM	bag	grey	fabric or textile	67.816 L	deposited

Logitech colors

Color	Seen	Deposited	Liters
grey	4	6	410.355 L



Intel RealSense D435

Measured stereo depth - factory intrinsics - hardware volume



BAGS SEEN
1

LIVE TRACKS
1

DEPOSITED
42

Logitech C920

Independent RGB silhouette tracking, colour, and material - monocular AI depth requires a measured reference distance

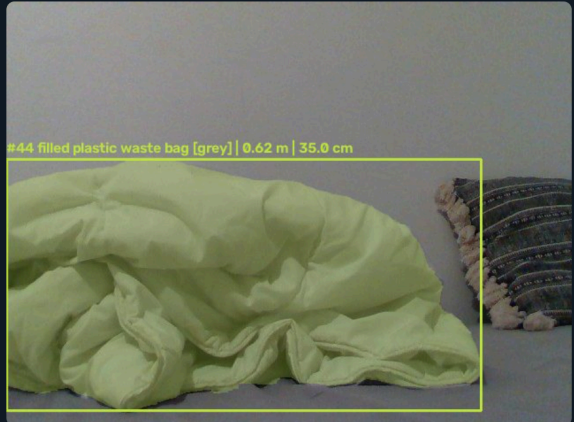


BAGS SEEN
1

LIVE TRACKS
1

DEPOSITED
11

Measured stereo depth - factory intrinsics - hardware volume



BAGS SEEN
5

LIVE TRACKS
1

DEPOSITED
42

CURRENT VOLUME
—

CUMULATIVE
562.877 L

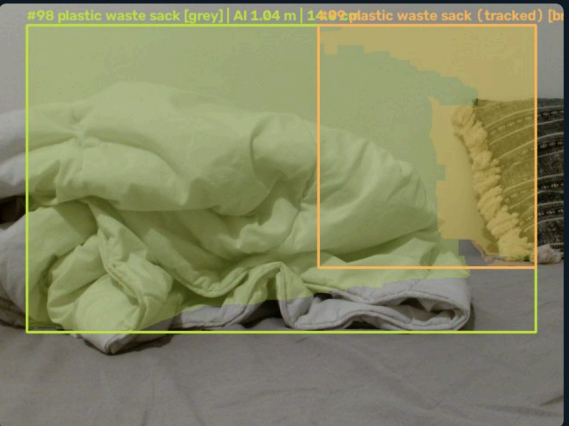
BIN OCCUPANCY
160.999 L

Capture baseline now

Reset baseline

Download RealSense CSV

Independent RGB silhouette tracking, colour, and material - monocular AI depth requires a measured reference distance



BAGS SEEN
10

LIVE TRACKS
2

DEPOSITED
11

CURRENT VOLUME
—

CUMULATIVE
643.603 L

BIN OCCUPANCY
160.543 L